The University of Montana

FORS 250 \ GPHY 284

Introduction to GIS and Cartography

Fall Semester 2024 Syllabus

FORS 250 \GPHY 284, Lecture Section 00, Rm SH 304
Tuesday and Thursday, 11:00-12:20 PM

FORS 250, Laboratory Section 01, Room SH 106
Friday, 11:00-12:50 PM

GPHY 284, Laboratory Section 02, Room SH 107
Friday, 12:00-1:50 PM

FORS 250, Laboratory Section 03, Rm SH 106
Friday, 1:00-2:50 PM

GPHY 284, Laboratory Section 04, Rm SH 107
Friday, 2:00-3:50 PM

FORS 250, Laboratory Section 05, Rm SH 106
Friday, 3:00-4:50 PM

Instructor: Rebekah Fields
Office: Room 313, Natural Sciences (NS)
Office Telephone:
Office Hours: Tuesday/Thursday 9:00 AM – 10:50, MWF – by appointment
Email: rebekah.fields@mso.umt.edu

TA: Colleen Livingstone
Office: TBA
Office Hrs: TBA
Email: colleen.livingstone@umconnect.umt.edu

TA: Bryan Tutt
Office: TBA
Office Hrs: TBA
Email: bryan.tutt@umconnect.umt.edu

COURSE DESCRIPTION:

This course is designed as a practical introduction to the use of Geographic Information Systems (GIS) for storing, retrieving, analyzing and displaying spatial data. It will also cover the history of cartography as an art form and the cartographic conventions of the modern map-making process. Finally, we will review many examples of applied GIS in the private, public, and non-profit sectors.

The course will discuss all the major concepts and theories behind the use of GIS and put these skills into practice with laboratory exercises using GIS software. We will closely examine the proper use of data, both from its appropriate choice given a desired outcome, and to inform a deeper understanding of data manipulation within the "Black Box." Secondly, the course will address the concepts of proper map design so that our maps more effectively convey the information we intend to impart.

Objectives:

- To learn the foundations of GIS in order to properly and truthfully display spatial data in a cartographic context.
- To understand the structure of spatial data including file associations, attribute tables, metadata, coordinate systems, and projections.
- To become proficient in the construction of geodatabases (GDB) as the fundamental data storage format within all GIS Science.
- To learn basic GIS analysis for answering rudimentary spatial questions using the scientific method.
- To develop software skills in programs used for map production in the modern cartographic workflow.

Learning Outcomes *- By the end of this course you will*:

1. Understand the core concepts and functions of geographic information systems.
2. Be able to discriminate among different characterizations of the Earth's surface and choose projections that are suitable for specific forestry, geography, and natural resources applications.
3. Recognize and acquire different forms of spatial data, normalize its spatial reference systems, and organize it into a geodatabase (GDB) in preparation for GIS analysis.
4. Be familiar with techniques of spatial analysis applied in forestry, geography, and natural resource management.
5. Be able to produce maps that appropriately display the outcomes of spatial analysis and communicate the results using proper cartographic conventions.

Course Format:

The general program for each week will be lectures Tuesday and Thursday, followed by Friday lab sections. Lecture days will start with announcements and then be followed by a presentation on the principles of GIS and cartography. New labs will be introduced in the Thursday lecture, but will be done face-to-face in the classroom on Friday. I will post the pdfs of the lecture Power Points to Canvas before each lecture.

Required Storage: A thumb-drive or external hard drive with at least 4 GB of space if you plan to work on campus. If working from home, you will need at least 2 GB of free space on your computer for all the materials in this class.

Server Address: T:\classes$\

Required Textbook:

Bolstad, Paul & Manson, Steven (2022). GIS Fundamentals, 7th Edition
ISBN: 9780971764750

Online text available at RedShelf or Vital Source

An online access pass is $18.00 for 180 days, or lifetime access for $24.00. You do not need to purchase the downloadable or print version, however those are available here: https://shop.btpubservices.com/Title/9780971764750 for $44.00 (or other book retailers).

POLICIES AND PROCEDURES:

The following policies are the minimum standards for which all students are responsible. They set the ground rules so that class can move forward in an efficient and productive manner. Please review and put into practice:

- Please consult the Class Schedule for relevant dates. All assignments will require submission on the due date specified in the Class Schedule unless otherwise noted.

- See the Deadlines Calendar for registration changes or withdrawals.

- **ATTENDANCE:** We all know external life events can and will affect attendance. Things like sicknesses or family emergencies may come up during the semester, attendance does not count towards your grade. **HOWEVER**, attendance IS incentivized. This means in-class "events" are randomly applied through the semester that add to your grade, events like extra credit opportunities, additional points applied to tests, and rounded up grades at the end of the semester.

- Attending class benefits you. All powerpoints are posted on Canvas the day before lecture, but they won't be particularly useful unless you're in class. Should you need to miss a class, you are responsible for keeping up with the material either by reading the book or getting notes from another student.

- **LATE WORK:** Life happens. Everyone gets ONE free, no questions asked pass for late work. Each day late for assignments is 5% off for a maximum of 50% (10 days late). After that, no late work will be accepted (except under special circumstances).

- **TEST CORRECTIONS:** You have the opportunity to correct your midterm test grade for half credit. If you get 50% on the test, you can correct your answers for a return of 25%, raising your grade to 75%. You must explain your corrected answer for credit, and it must be returned one week after the test.

- **AI POLICY**: As technology continues to advance, Artificial Intelligence (AI) has moved to the forefront of conversations surrounding learning, assessments, and career training. I have found many AI Models helpful in my own writing and coding. It would be disingenuous of me to restrict a resource that I use myself. That being said, we will explore the use of AI including in writing, searching for information, data development and more. HOWEVER, as members of our society, whatever career path you choose to take, you have a responsibility to use critical thinking to determine whether using AI will enhance or reduce your learning. This is an important topic and one that continues to evolve. I look forward to engaging with AI and your thoughts around it more this semester.

- **CLASSROOM ETIQUETTE**: A positive learning environment relies upon creating an atmosphere where diverse perspectives can be expressed, especially in today's political and social era. Each student is encouraged to take an active part in class discussions and activities. Honest and respectful dialogue is expected. Disagreement and challenging of ideas in a supportive and sensitive manner is encouraged. Hostility and disrespectful behavior is not acceptable. **However, in this class, our emphasis will be on engaging in the mutual exploration of issues as presented in the course as scholars rather than in defending points of view we have formed outside the classroom.** – Adapted from Evelyn Alsutany, University of Michigan

- **WRITING:** For assistance with writing, please consult the on-line resources of the UM Writing Center. The Writing Center provides one-on-one tutoring to students at all levels and at any time in the writing process. Visit now. Visit often.

- For assistance with MS Office, please consult this on-line resource.

- All students must practice academic honesty. Academic misconduct is subject to an academic penalty by the course instructor and/or a disciplinary sanction by the University. All students need to be familiar with the Student Conduct Code.

- Students with disabilities may request reasonable modifications by contacting the

disability office. The University of Montana assures equal access to instruction for students with disabilities in collaboration with instructors and the Office for Disability Equity. The University does not permit fundamental alterations of academic standards or retroactive modifications.

Canvas:

Class materials will be available on **Canvas** the week they are covered in class. This syllabus will be posted in the header for the entire semester. Refer to it often for due dates and reading assignments. Each lab will be submitted to its own digital drop box on **Canvas.** They will be graded with comments within the Canvas grade book.

LABS:

Throughout the semester we will be meeting weekly to practice our GIS skills using ESRI Software. Here are a few things to note:

While I try my best to write detailed instructions, it can be easy to get lost if you skip a section or miss a step. I know how frustrating that can be but I will ask you to try to solve the problem on your own or with a lab mate for at least *5 minutes*. The TA's and I will be around but in order for you to become fluent in the software you will need to learn how to use it on your own.
And look, sometimes problems pop up! We are still willing to help support you we just want you to try to solve the problem first before asking for help.

Each lab will be graded in two parts, a "quiz" on Canvas, and either a Map or other submission. Those deliverables will be described at the beginning of each lab.
The "quiz" on Canvas is not really a quiz, it's a series of questions you will need to answer, but you can use any resources at your disposal including lab mates, the maps you make and any googling you want. I also have the quizzes set up so you can take them twice, so if you struggle the first time, take it again and take the higher grade!
The other submissions will usually be a map that we pull from the software and paste into a word document. The Map instructions and points will *usually* be listed on each lab, but as the semester progresses, I expect you to become fluent in what factors maps need to be considered good maps.

ESRI software is only available on PC, no Macs here! That means navigating through a Windows environment may be a little more difficult for those of you who are used to Mac interfaces.

Labs can be completed in one of three ways:

1. Use the computer labs in person on campus. The instructor or TAs will be present to help with the labs every Friday (best and recommended).
2. Install a student license for the GIS software on your computer and work from home. This is a good option but requires you to have a newer PC (Sorry, I didn't make the software).
3. Remotely connect to the computer labs and use the software over the internet from home.

This is the best option for those with older computers or a Mac but requires a good internet connection (can be challenging to get set up).

4. Use the library computers to complete the labs (a good alternative to attending labs).

Instructions for how to download the software, remotely connect to the virtual computers, or access the computer labs in person will be provided on Canvas the first week of class. If you have technology barriers that will hinder you from completing the laboratory portion of this course, please contact Forestry IT: support@cfc.umt.edu. They have resources to help with computer-related issues.

I will say this many times through the semester: If you have *any* barriers to completing labs, please contact me or a TA. We are here to support you, and we cannot help you if you don't tell us what you need! Additionally, reasonable accommodations will be made for special circumstances but are difficult to apply retroactively. Let us know ASAP if you know of anything you may be facing.

CLASS ASSIGNMENTS:

- ***Lab Exercises***
  Labs will consist of exercises that provide a means to put theory, as presented through the lectures and reading material, into practice. The labs are software intensive utilizing ESRI ArcPro, and Microsoft Excel or Google Sheets. There will be instructions posted to Canvas for each lab. Each lab will have a required assignment that will test your knowledge and use of the software. These assignments will be due Friday the following week at midnight, and late labs will be accepted for one more week for a 30% deduction. All assignments will be submitted on Canvas.

- ***Lab Final***
  You will have a final lab project for this class worth 100 pts. Details for the assignment will be given in class and you will have lab time to work on the project.

- ***In class quiz***
  There will be one, open note, in class quiz going over the first three weeks of class worth 30 points. Why? Because these are fundamental concepts we will be using for the rest of the semester, and you will need to have a good understanding of these concepts to be successful in this and future GIS classes.

- ***Midterm and Final Exams***
  Exams will be administered in person, and will include topics from lectures, labs and readings. The final exam **IS Comprehensive**, however, it will mostly consist of material covered in the second half of the semester. The Final Exam is longer and will be worth more than the Midterm. See the Points Table below for details.

- ***Extra Credit Assignments***

There will be several opportunities for outside coursework that counts as extra credit. Two available to all students are described below, while others will be bonuses as described above.

- Participate in the 30-day map challenge:
  - The 30-day map challenge was a challenge started on twitter in 2019, where for the entire month of November cartographers around the world create unique maps based on the topic of that day. To receive up to10 points of extra credit, complete 5 of the days' categories in whatever way you choose. Make sure to adhere to the rules of maps to be discussed in lab. See Canvas for more information.
- Make a map!
  - Make a map of anything you want, outside of a computer software. This can be any material and size, but must follow the rules of maps we discussed in lab. See Canvas for more information.

If you are having trouble with a lab, contact a TA or myself well before it is due. If you have an emergency, illness, or crisis: send an email or call my office before the assignment is due.

GRADING:

This 3-credit class is offered for traditional letter grade only (T). It is not offered under the credit/no credit option. The tables below break down the point values for all required assignments. Grades are evaluated on the completeness and organization of the project, as well as the use of the theory and techniques taught in class. Maps will not be graded purely on a subjective assessment of aesthetic appeal; however, a poorly executed map is certainly worth less than a professional one. Not everyone is an artist, but the student should demonstrate progress toward GIS and cartographic competency.

All assignments, as well as the final grade, are evaluated on the following grading scale:

A ..... 95 – 100%
A- …... 90 – 94.99%
B+ ..... 87 – 89.99%
B ..... 83 – 86.99%
B- ..... 80 – 82.99%
C+ ..... 77 – 79.99%
C ..... 73 – 76.99%
C- ..... 70 – 72.99%
D ..... 60 – 69.99%
F ..... 59.99% and below

***Please note** the lowest lab grade will be dropped, and you have the chance to correct your midterm exam for half credit.

Points Table:

Assignments	**Points**
Lab 1	20 pts.
Lab 2	20 pts.
Lab 3	20 pts.
Lab 4	20 pts.
Lab 5	30 pts.
Lab 6	35 pts.
Lab 7	65 pts.
Lab 8	50 pts.
Lab 9	75 pts.
Lab 10	75 pts.
Lab Final	100 pts.
In class quiz	30 pts.
Thematic Cartography Assignment	30 pts.
Midterm Exam	140 pts.
Final Exam	140 pts.
Total	**820 pts.**

Class Schedule			
Week 1: Intro to GIS and Cartography			
Week/Day	**Topic**	**Assignment**	**Reading (book pages in bold)**
Tuesday 8/27	Welcome, Syllabus, Canvas, Labs, Resources	Video	Page 13-17
Thursday 8/29	What is GIS? History of Cartography		pg 37-40 (stop at Coordinate System); 48 (start at Attribute Data and Types) – 58 **pg 25-28; 36-46**
Friday 8/30	Lab 1: Intro to Arc Pro	**Lab 1 due Friday 9/6**	
Week 2: The structure of Geospatial Data			
Tuesday 9/3	The Geodatabase and Data Types		pg 161 – 164 **pg 149 - 152**
Thursday 9/5	Geodesy, Coordinate Systems		pg 101- 106 (stop at "Most modern datums…"); 117- 119. **pg 89-94; 105-107**
Friday 9/6	Lab 2: Building Geodatabases, Importing data	**Lab 2 due Friday 9/13**	
Week 3: Shape and scale: the map compromise			
Week/Day	**Topic**	**Assignment**	**Reading**
Tuesday 9/10	Projections and scale		
Thursday 9/12	Refresh and review	**In class quiz**	
Friday 9/13	Lab 3: Projections	**Lab 3 due Friday 9/20**	
Week 4: Cartographic principles			
Tuesday 9/17	Map labels and layout	Good Mapmaking PDF, Make Maps People Want to Look at (Canvas)	pg. 78, 485-493 **pg. 66, 473-481**
Thursday 9/19	Vector analysis and symbolism	**https://www.youtube.com/watch?v=fvzNkdmoy48**	pg 485-488 **pg 473-476**
Friday 9/20	Lab 4: Coordinate Systems and Data Management	**Lab 4 due Friday 9/27**	

Week 5: Raster data and the earth from space			
Tuesday 9/24	Digital Elevation Models (DEM) & Hillshading		pg 251 – 280, 289- 291 **pg. 239-268, 277- 279**
Thursday 9/26	Satellite Imagery	https://radiolab.org/podcast/eye-sky	
Friday 9/27	Lab 5: Creating a Shaded Relief Topographic Map	**Lab 5 due Friday 10/4**	
Week 6: Vector data – points, lines and polygons			
Week/Day	**Topic**	**Assignment**	**Reading**
Tuesday 10/1	GPS	**Exam review**	
Thursday 10/3	Exam Review	**Exam review – mind map/note sheet**	
Friday 10/4	Lab 6: GPS Mapping	**Lab 6 due Friday 10/11**	
Week 7: Midterm week			
Tuesday 10/8	**Midterm**		
Thursday 10/10	Intro to Arc Online	Video – this video is in prep for lab, please watch it before attending.	
Friday 10/11	Lab 7: Creating and Using Survey123		
Week 8: Arc Online I			
Tuesday 10/15	TBD		
Thursday 10/17	TBD		
Friday 10/18	Continue Lab 7	**Lab 7 due Friday 10/25**	
Week 9: Thematic Maps			
Week/Day	**Topic**	**Assignment**	**Reading**
Tuesday 10/22	Choropleth and Graduated Symbols Maps	https://academy.datawrapper.de/article/134-what-to-consider-when-creating-choropleth-maps	

Thursday 10/24	Isarithmic and Dot Density Maps	Thematic Cartography Assignment due 10/31	
Friday 10/25	Lab 8: Choropleth Maps of Population		
Week 10: Advanced GIS			
Tuesday 10/29	Spatial Modeling and Models		
Thursday 10/31	Raster Analysis		
Friday 11/1	Lab 9: Forest Management	**Lab 9 due Friday 11/8**	
Week 11: Applied GIS I			
Tuesday 11/5	GIS for topics in resource management	https://99percentinvisible.org/episode/trail-mix/	
Thursday 11/7	GIS for topics in social sciences		
Friday 11/8	Lab 10: Recreation management		
Week 12: Applied GIS II			
Tuesday 11/12	Introduce Lab Final		
Thursday 11/14	*No class*	**Get project idea approved**	
Friday 11/15	Lab Final Working day	**Lab final due 12/6**	
Week 13:			
Tuesday 11/19			
Thursday 11/21			
Friday 11/22	Review		
Week 14: Thanksgiving week; NO CLASS			
Week 15: Finals Week			
Tuesday 12/3	Review		
Thursday 12/5	Final		

Friday 12/6	Lab Final Due		
FINALS WEEK			